Yashasvi Karnena

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Nationality: Indian
Date of Birth: 24.05.1997

Professional Experience

Nov. 2021 – Present | **fka GmbH** – Aachen, Germany

Development Engineer – Embedded Software (Apr. 2023 – Present)

- Developed ISO-26262-qualified real-time control software components using TargetLink and MATLAB/Simulink.
- Built Functional Mockup Units (FMUs) for validating ECU firmware against controlled plant models.
- Investigated Nonlinear control methods for PMSM position control in Steer-By-Wire systems.
- Conducted HiL, MiL, and SiL testing of functional software.
- Researched and implemented concepts for cloud based plant parameter estimation and adaptive controller tuning of Software Defined Vehicles.

Master Thesis Student (Apr. 2022 – Mar. 2023)

- Implemented position trajectory tracking controller using nonlinear 2DOF control.
- Developed nonlinear MPC trajectory planner with State-Lattice planners for solver warm-starting to tackle non-convexities.
- Devised a state-machine based sequential motion cooperation algorithm for Connected Autonomous Vehicles.

Intern (Nov. 2021 – Mar. 2022)

- Designed cascaded position control loops and extended Kalman filters (Parameter Estimation) for Steer-by-Wire systems.
- Developed C/C++ estimators and filters; deployed on PXROS RTOS.
- Debugged Hardware, Firmware and control loop problems.

May 2021 – Oct. 2021 | **Institute for Electrical Machines, RWTH Aachen** – Aachen, Germany

Student Assistant

- Created Simulink exercises for lab course *Dynamics of Electrical Machines*.
- Implemented Simulink based bearing failure analysis models arising out of inverter harmonics.

May 2018 – Apr. 2019 | **Altigreen Propulsion Labs Private Limited** – Bengaluru, India

Junior Embedded Engineer

- Built firmware for telematics units and Inverter FOC algorithms in Simulink.
- Set up toolchains using CMake/Jenkins for TI C2000 and ARM platforms.

Apr. 2017 – Sept. 2017 | **Defense Avionics Research Establishment** – Bengaluru, India

Electrical Engineering Intern

- Designed power distribution boards using LTSpice and Altium.
- Implemented avionics logic on FPGAs; developed analog/digital filters.

Education

Oct. 2019 – May 2023 | **RWTH Aachen University** – Aachen, Germany
M.Sc. in Electrical Engineering, IT and Computer Engineering
Major: Systems and Automation (Grade: 2.2 / 4.0)

Jul. 2014 – May 2018 | **Vellore Institute of Technology** – Vellore, India
B.Tech. in Electrical and Electronics Engineering
(Grade: 8.36 / 10)

Skills

Programming: C/C++, Java, Python, Golang, Bash, LaTeX, CMake
Simulation: MATLAB, Simulink, LabView, PLECS, PSpice, TargetLink
EDA Tools: KiCAD, Altium Designer, Eagle PCB
Testing: CANoe, CANape, ControlDesk, PCAN-View
IT Tools: Git, Jira, MS Office, IBM DOORS, Windchill, Citavi
Languages: English (C2), German (B1), Telugu (C2), Hindi (C1)

Certifications

Oct. 2024 | **Automotive Cyber Security Training** – NobleProg GmbH

Extracurricular Activities

Apr. 2022 – Sept. 2024 | **Tachyon Hyperloop** – RWTH Aachen University
- Implemented Telemetry, Brakes Control and, Thermal Control hardware using Altium Designer.
- Developed high voltage power electronics package for 3-Phase propulsion control.
- Developed firmware using MATLAB/Simulink for TI C2000 microcontrollers.
- Managed finances and projects as a board member and mentored new recruits in electronics design.

Feb. 2015 – Sept. 2017 | **Pravega Racing** – Vellore Institute of Technology
- Developed Electronic Throttle controller.
- Tuned Engine Control Units and built race specification wiring harness.
- Developed Traction Control Unit for race car launch control.

Awards

Aug. 2024 | **European Hyperloop Week** – Best Electronics Package

Jan. 2017 | **Formula Student India** – Best Engineering Design

May 2013 | **NASA Space Settlement Contest** – Honorable Mention

Yashasvi Karnena
Aachen, September 2, 2025